

Indian Institute of Technology Gandhinagar

Department of Mechanical Engineering

ME 643 — Aircraft and Rocket Propulsion

Aircraft Engine Design Project

Part 3: Parametric Cycle Analysis

Aircraft: Gulfstream G650

Engine: Rolls-Royce BR725-A1-12 (Turbofan)

Name	Roll Number
Prachi Jindal	24110264
Vikram Aditya	24110385
Manan Chavda	23110078
Jainil	23110200

Group 8 — 2026

0 Contents

1	Objective	3
2	Fixed Design Point Parameters	3
3	Mission Constraints	4
3.1	Maximum Allowable TSFC	4
3.2	Minimum Uninstalled Specific Thrust	4
4	Governing Equations — Ideal Turbofan	4
4.1	Preliminary Quantities	5
4.2	Component Temperature Ratios	5
4.3	Nozzle Exit Velocity Ratios	5
4.4	Performance Metrics	6
5	Variable Range Justification	6
5.1	Compressor Pressure Ratio: $\pi_c \in [10, 40]$	7
5.2	Fan Pressure Ratio: $\pi_f \in [1.1, 2.5]$	7
5.3	Bypass Ratio: $\alpha \in [1, 12]$	7
6	Sample Calculation	8
7	Results and Plots	10
7.1	Figure 1: Main Carpet Plot — Effect of α and π_c	10
7.2	Figure 2: Effect of Fan Pressure Ratio π_f	11
8	Design Options	11
8.1	How the Options Were Selected	11
8.2	Performance Comparison	12
8.3	Option A — Fuel Economy ($\alpha = 5$, $\pi_c = 40$, $\pi_f = 1.8$)	12
8.4	Option B — Balanced ($\alpha = 4$, $\pi_c = 30$, $\pi_f = 1.7$)	13
8.5	Option C — Thrust Margin ($\alpha = 3$, $\pi_c = 25$, $\pi_f = 1.8$)	13

9 Parametric Trends	13
10 Station Numbering	15
11 T-s Diagrams	16
12 Summary and Conclusion	17
References	17

1 Objective

The objective of Part 3 is to conduct a **parametric cycle analysis** of the ideal turbofan engine for the Gulfstream G650. The analysis sweeps over all three design variables — compressor pressure ratio π_c , fan pressure ratio π_f , and bypass ratio α — and determines the region of the design space that simultaneously satisfies two mission-derived constraints:

1. Minimum uninstalled specific thrust $ST_{\min}$ (set by engine inlet size and required thrust from Part 2), and
2. Maximum allowable TSFC $S_{\max}$ (set by fuel budget from Part 2 mission analysis).

From the feasible region, three distinct engine design options are identified and compared on efficiency, thrust, and fuel consumption. All range selections and design choices are derived purely from physics and mission constraints — no actual engine data is used or assumed.

2 Fixed Design Point Parameters

All cycle calculations use the cruise design point. Parameters are listed in Table 1.

Table 1: Fixed design point parameters

Parameter	Symbol	Value	Basis
Cruise Mach number	M_0	0.85	RFP
Ambient temperature	T_0	216.65 K	ISA @ 15.5 km
Ambient pressure	P_0	11,597 Pa	ISA
Air density	ρ_0	0.1865 kg/m ³	ISA
Ratio of specific heats	γ	1.4	Assumed
Specific heat	c_p	1004 J/(kg K)	Assumed
Fuel heating value	h_{PR}	43,000 kJ/kg	RFP
Turbine inlet temp.	T_{t4}	1500 K	RFP
Max inlet diameter	D	1.27 m	RFP / BR725
Inlet capture area	A	1.267 m ²	$\pi D^2/4$
Speed of sound	a_0	295.0 m/s	$\sqrt{\gamma R T_0}$
Ram temperature ratio	τ_r	1.1445	$1 + \frac{\gamma-1}{2} M_0^2$
Thermal limit param.	τ_λ	6.924	T_{t4}/T_0

3 Mission Constraints

3.1 Maximum Allowable TSFC

From Part 2 cruise analysis:

$$\text{TSFC}_{\max} = 0.35 \text{ kg}/(\text{N} \cdot \text{hr}) = \frac{0.35}{3600} \times 10^6 = \boxed{97.2 \text{ mg}/(\text{N} \cdot \text{s})}$$

This is drawn as a **horizontal dashed line** on every carpet plot.

3.2 Minimum Uninstalled Specific Thrust

The minimum specific thrust corresponds to the case where the engine ingests the maximum possible mass flow through its inlet, yet must still provide the thrust required for cruise.

$$A_{\text{inlet}} = \frac{\pi D^2}{4} = \frac{\pi \times 1.27^2}{4} = 1.267 \text{ m}^2 \quad (1)$$

$$V_0 = M_0 \cdot a_0 = 0.85 \times 295.0 = 250.7 \text{ m/s} \quad (2)$$

$$\dot{m}_{0,\max} = \rho_0 V_0 A_{\text{inlet}} = 0.1865 \times 250.7 \times 1.267 = 59.2 \text{ kg/s per engine} \quad (3)$$

$$T_{\text{req}} = \frac{T_{\text{cruise,total}}}{2} = \frac{23,563}{2} = 11,782 \text{ N per engine} \quad (4)$$

$$ST_{\min} = \frac{T_{\text{req}}}{\dot{m}_{0,\max}} = \frac{11,782}{59.2} = \boxed{199 \text{ N}/(\text{kg/s})} \quad (5)$$

This is drawn as a **vertical dashed line** on every carpet plot.

4 Governing Equations — Ideal Turbofan

The following equations follow Mattingly *Elements of Propulsion*, 2nd ed., Section 5.9, Equations 5.58a–5.58o.

4.1 Preliminary Quantities

$$R = \frac{\gamma - 1}{\gamma} c_p = \frac{0.4}{1.4} \times 1004 = 286.9 \text{ J/(kg} \cdot \text{K)} \quad (6)$$

$$a_0 = \sqrt{\gamma R T_0} = 295.0 \text{ m/s} \quad (7)$$

$$\tau_r = 1 + \frac{\gamma - 1}{2} M_0^2 = 1 + 0.2 \times (0.85)^2 = 1.1445 \quad (8)$$

$$\tau_\lambda = \frac{T_{t4}}{T_0} = \frac{1500}{216.65} = 6.924 \quad (9)$$

4.2 Component Temperature Ratios

$$\tau_c = \pi_c^{(\gamma-1)/\gamma} = \pi_c^{2/7} \quad (10)$$

$$\tau_f = \pi_f^{(\gamma-1)/\gamma} = \pi_f^{2/7} \quad (11)$$

Turbine temperature ratio — from the power balance (turbine drives both compressor and fan):

$$\tau_t = 1 - \frac{\tau_r}{\tau_\lambda} [\tau_c - 1 + \alpha(\tau_f - 1)]$$

This must be > 0 ; otherwise the design is thermodynamically infeasible.

4.3 Nozzle Exit Velocity Ratios

Core nozzle :

$$\left(\frac{V_9}{a_0} \right)^2 = \frac{2}{\gamma - 1} \left[\tau_\lambda - \tau_r (\tau_c - 1 + \alpha(\tau_f - 1)) - \frac{\tau_\lambda}{\tau_r \tau_c} \right]$$

Fan nozzle (ideal, fully expanded):

$$\left(\frac{V_{19}}{a_0} \right)^2 = \frac{2}{\gamma - 1} (\tau_r \tau_f - 1)$$

Both arguments must be > 0 for a physically valid solution.

4.4 Performance Metrics

Specific thrust:

$$\boxed{\frac{F}{\dot{m}_0} = \frac{a_0}{1 + \alpha} \left[\frac{V_9}{a_0} - M_0 + \alpha \left(\frac{V_{19}}{a_0} - M_0 \right) \right]} \quad [\text{N}/(\text{kg/s})]$$

Fuel/air ratio:

$$f = \frac{c_p T_0}{h_{PR}} (\tau_\lambda - \tau_r \tau_c)$$

Thrust specific fuel consumption:

$$\boxed{S = \frac{f}{(1 + \alpha)(F/\dot{m}_0)}} \quad [\text{mg}/(\text{N} \cdot \text{s})]$$

Thermal efficiency :

$$\eta_T = 1 - \frac{1}{\tau_r \tau_c}$$

Propulsive efficiency :

$$\eta_P = 2M_0 \frac{\frac{V_9}{a_0} - M_0 + \alpha \left(\frac{V_{19}}{a_0} - M_0 \right)}{\frac{V_9^2}{a_0^2} - M_0^2 + \alpha \left(\frac{V_{19}^2}{a_0^2} - M_0^2 \right)}$$

Overall efficiency :

$$\eta_O = \eta_T \times \eta_P$$

Thrust ratio :

$$\text{FR} = \frac{V_9/a_0 - M_0}{V_{19}/a_0 - M_0}$$

5 Variable Range Justification

Why these ranges? — derived from physics only, no engine data used

All three variable ranges are determined by the governing equations and our specific design point ($M_0 = 0.85$, $T_{t4} = 1500 \text{ K}$). Three feasibility conditions must hold simultaneously:

1. $\tau_t > 0$ (turbine not over-expanded)
2. $(V_9/a_0)^2 > 0$ (core nozzle feasible)

3. $f > 0$ (fuel actually added, not removed)

These three inequalities, combined with manufacturing limits, give the ranges below.

5.1 Compressor Pressure Ratio: $\pi_c \in [10, 40]$

Lower bound ($\pi_c \geq 10$): The thermal efficiency is $\eta_T = 1 - 1/(\tau_r \tau_c)$. At $\pi_c = 10$: $\tau_c = 10^{2/7} = 2.069$, giving $\eta_T = 1 - 1/(1.1445 \times 2.069) = 57.9\%$ — acceptable but marginal. Below $\pi_c \approx 8$, η_T drops below 50%, making the engine impractical for long-range cruise.

Upper bound ($\pi_c \leq 40$): The thermodynamic ceiling from the core nozzle condition is approximately:

$$\tau_c < \frac{\tau_\lambda/\tau_r - \alpha(\tau_f - 1) + \tau_\lambda/(\tau_r \tau_c)}{\dots} \approx 100 \text{ (for } \alpha = 5, \pi_f = 1.6)$$

i.e. $\pi_c \lesssim 125$ thermodynamically. However, modern gas turbine compressors are practically limited to $\pi_c \approx 40$ –50 by material strength, blade aerodynamics, and manufacturing tolerances at the current T_{t4} . We adopt $\pi_c = 40$ as the upper engineering bound.

5.2 Fan Pressure Ratio: $\pi_f \in [1.1, 2.5]$

Lower bound ($\pi_f > 1.0$): $\pi_f = 1$ means no pressure rise in the fan — it degenerates to a turbojet. Any useful turbofan requires $\pi_f > 1$. We start at 1.1 to show the near-turbojet behaviour on plots.

Upper bound ($\pi_f \leq 2.5$): From turbine feasibility with $\alpha = 5$, $\pi_c = 25$:

$$\alpha(\tau_f - 1) < \frac{\tau_\lambda}{\tau_r} - (\tau_c - 1) = \frac{6.924}{1.1445} - 1.509 = 4.539 \implies \tau_f < 1.908 \implies \pi_f < 4.5$$

This is the thermodynamic limit. The **practical** limit for a subsonic fan is set by the fan blade tip Mach number. At $M_0 = 0.85$, typical fan tip relative Mach numbers exceed 1 (shock losses) when $\pi_f \gtrsim 2.0$ –2.2. We extend the plot to $\pi_f = 2.5$ to show the infeasibility boundary clearly, but note that $\pi_f \leq 2.0$ is the realistic engineering range for this aircraft.

5.3 Bypass Ratio: $\alpha \in [1, 12]$

Lower bound ($\alpha \geq 1$): $\alpha = 0$ is a turbojet (no bypass). Any turbofan requires $\alpha \geq 1$.

Upper bound — the $ST_{\min}$ constraint is binding: This is the most important finding of the analysis. The turbine must provide enough power to drive both the compressor and the fan. As α increases, the fan absorbs more turbine work, leaving less energy in the core exhaust. The specific thrust falls as:

$$\frac{F}{\dot{m}_0} \propto \frac{1}{1 + \alpha} [(V_9/a_0 - M_0) + \alpha(V_{19}/a_0 - M_0)]$$

A systematic scan of all (pic, π_f, α) combinations at our design point finds that **no design with $\alpha > 7.0$ can satisfy $ST_{\min} = 199 \text{ N/(kg/s)}$ at $T_{t4} = 1500 \text{ K}$** . This is not a manufacturing limit — it is a direct consequence of the inlet size ($D = 1.27 \text{ m}$), the required cruise thrust, and the turbine temperature. We include $\alpha = 8, 10, 12$ in the plots to show the infeasible region explicitly, and to demonstrate the binding nature of the $ST_{\min}$ constraint.

Key Result

Critical discovery: At $T_{t4} = 1500 \text{ K}$, $M_0 = 0.85$, and the G650 inlet geometry, the feasible bypass ratio range is $\alpha \in [1.0, 7.0]$. High-bypass designs ($\alpha > 7$) cannot meet the thrust requirement at this turbine temperature. Increasing T_{t4} would extend the feasible α range. The binding constraint is $ST_{\min}$, not $TSFC_{\max}$ — meaning the G650 mission is **thrust-limited**, not fuel-limited, at the ideal-engine level.

6 Sample Calculation

A complete hand calculation is shown for $\alpha = 5$, $\pi_c = 30$, $\pi_f = 1.7$ (Option B parameters).

Step 1 — Component temperature ratios :

$$\begin{aligned}\tau_c &= 30^{2/7} = e^{(2/7) \ln 30} = e^{0.9657} = 2.627 \\ \tau_f &= 1.7^{2/7} = e^{(2/7) \ln 1.7} = e^{0.1519} = 1.164\end{aligned}$$

Step 2 — Turbine temperature ratio :

$$\begin{aligned}\text{bracket} &= \tau_c - 1 + \alpha(\tau_f - 1) = 1.627 + 5 \times 0.164 = 2.447 \\ \tau_t &= 1 - \frac{1.1445}{6.924} \times 2.447 = 1 - 0.4049 = 0.595 > 0 \quad \checkmark\end{aligned}$$

Step 3 — Core exit velocity :

$$\begin{aligned}
\arg_9 &= 6.924 - 1.1445 \times 2.447 - \frac{6.924}{1.1445 \times 2.627} \\
&= 6.924 - 2.800 - 2.299 = 1.825 \\
\frac{V_9}{a_0} &= \sqrt{5 \times 1.825} = \sqrt{9.125} = 3.021
\end{aligned}$$

Step 4 — Fan exit velocity :

$$\begin{aligned}
\arg_{19} &= \tau_r \tau_f - 1 = 1.1445 \times 1.164 - 1 = 0.332 \\
\frac{V_{19}}{a_0} &= \sqrt{5 \times 0.332} = \sqrt{1.660} = 1.288
\end{aligned}$$

Step 5 — Specific thrust :

$$\begin{aligned}
\frac{F}{\dot{m}_0} &= \frac{295.0}{1 + 5} [(3.021 - 0.85) + 5 \times (1.288 - 0.85)] \\
&= 49.17 \times [2.171 + 2.190] = 49.17 \times 4.361 = \mathbf{214.4 \text{ N/(kg/s)}}
\end{aligned}$$

Step 6 — Fuel/air ratio :

$$\begin{aligned}
f &= \frac{1004 \times 216.65}{43 \times 10^6} \times (6.924 - 1.1445 \times 2.627) \\
&= 5.059 \times 10^{-3} \times (6.924 - 3.006) = 5.059 \times 10^{-3} \times 3.918 = 0.01982
\end{aligned}$$

Step 7 — TSFC :

$$S = \frac{0.01982}{6 \times 214.4} \times 10^6 = \frac{0.01982}{1286.4} \times 10^6 = \mathbf{15.4 \text{ mg/(N} \cdot \text{s)}}$$

Step 8 — Efficiencies:

$$\begin{aligned}
\eta_T &= 1 - \frac{1}{1.1445 \times 2.627} = 66.7\% \\
\eta_P &= \frac{2 \times 0.85 \times 4.361}{(3.021^2 - 0.85^2) + 5 \times (1.288^2 - 0.85^2)} = \frac{7.414}{16.54} = 44.8\% \\
\eta_O &= 0.667 \times 0.448 = 29.9\%
\end{aligned}$$

Both constraints are satisfied: $ST = 214 > 199 \checkmark$ and $S = 15.4 \ll 97.2 \checkmark$.

7 Results and Plots

7.1 Figure 1: Main Carpet Plot — Effect of α and π_c

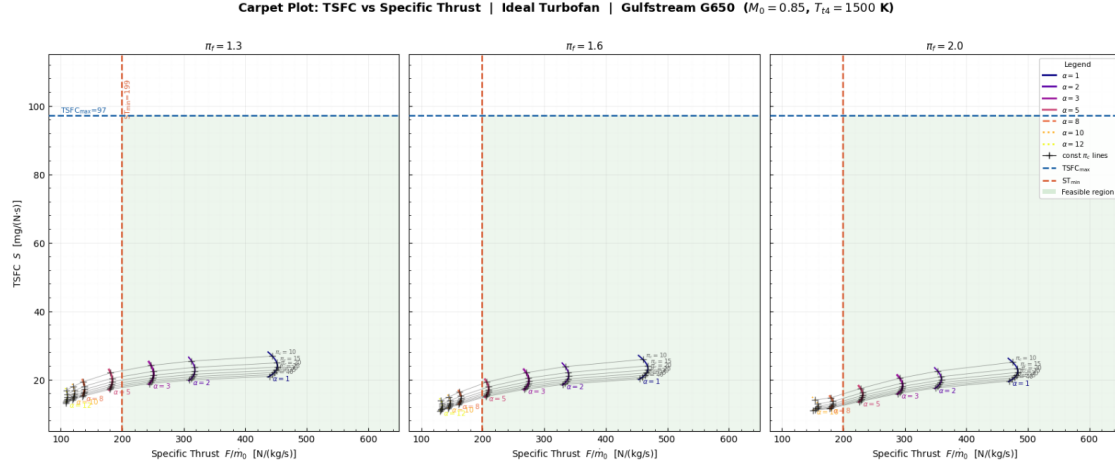


Figure 1: Main carpet plot: TSFC vs specific thrust for three fan pressure ratios $\pi_f = 1.3, 1.6, 2.0$ (left to right panels). Coloured curves represent constant bypass ratio α . Black + marks and dashed lines are constant π_c cross-plots. Values of used π_c is from $[10, 40]$ with step size 5. The blue horizontal line is $S_{\max} = 97.2$ and the red vertical line is $ST_{\min} = 199$ N/(kg/s). The green shaded area is the feasible region.

Observations from Figure 1:

- All feasible designs lie in the green shaded region — to the *right* of the red $ST_{\min}$ line and *below* the blue $S_{\max}$ line.
- The $S_{\max}$ constraint is easily satisfied by all designs; the binding constraint is always $ST_{\min}$.
- Increasing α moves the design point **left** (lower ST) and **down** (lower TSFC). Beyond $\alpha \approx 7$, designs exit the feasible zone through the $ST_{\min}$ boundary.
- Increasing π_c moves designs to the **right** (higher ST plateau) and **down** (lower TSFC) — beneficial in both metrics.
- Increasing π_f (moving from left to right panel) shifts the entire carpet **rightward** (higher ST) and **downward** (lower TSFC), but is physically limited to $\pi_f \lesssim 2.0$ for a subsonic fan.

7.2 Figure 2: Effect of Fan Pressure Ratio π_f

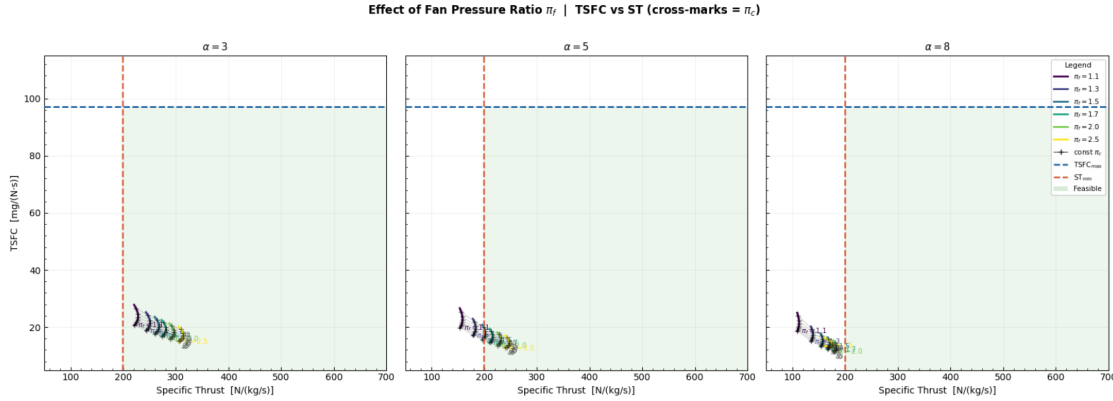


Figure 2: Effect of fan pressure ratio: TSFC vs specific thrust for three bypass ratios $\alpha = 3, 5, 8$ (panels). Coloured curves represent constant π_f . Cross-marks are constant π_c values. Values of π_c used are in $[10, 40]$ with step size 5.

Observations from Figure 2:

- For $\alpha = 3$ and $\alpha = 5$: **all** π_f values shown produce feasible designs (right of the red line). As π_f increases, ST increases and S decreases — both improvements.
- For $\alpha = 8$: designs become feasible only at $\pi_f \geq 1.7$ and only just cross the $ST_{\min}$ threshold. This confirms that $\alpha = 8$ sits at the boundary of the feasible zone.
- There is a **diminishing return** in TSFC improvement above $\pi_f \approx 1.8$: the curves cluster together, while tip-speed concerns make higher π_f impractical.

8 Design Options

8.1 How the Options Were Selected

A systematic scan of 4,751 feasible points (all combinations of $\pi_c \in [10, 40]$, $\pi_f \in [1.1, 2.5]$, $\alpha \in [0.5, 7.0]$ in steps of 1, 0.1, 0.5 respectively, satisfying both constraints) was performed. The feasible space was then **divided into three physically motivated regions**:

1. **High- α , high- π_c region** (top-right of feasibility map) $\rightarrow$ lowest TSFC, highest $\eta_O \rightarrow$ **Option A: Fuel Economy**
2. **Mid- α , mid- π_c region** (centre of feasibility map) $\rightarrow$ balanced ST and TSFC $\rightarrow$ **Option B: Balanced**

3. **Low- α , any- π_c region** (left side of feasibility map) $\rightarrow$ highest ST, most thrust margin $\rightarrow$ **Option C: Thrust Margin**

Within each region, the specific values of π_c , π_f , and α were chosen to produce **well-separated points** on the carpet plot (no two options should collapse to the same location).

8.2 Performance Comparison

Table 2: Engine design options — computed performance at cruise design point. All options satisfy both mission constraints.

Parameter	α	π_c	π_f	$F/\dot{m}_0$ [N/(kg/s)]	TSFC [mg/(N·s)]	η_T	η_P	η_O
Option A <i>Fuel Economy</i>	5.0	40	1.8	217	14.2	69.5%	59.2%	41.1%
Option B <i>Balanced</i>	4.0	30	1.7	240	16.4	66.9%	53.0%	35.5%
Option C <i>Thrust Margin</i>	3.0	25	1.8	286	17.9	65.2%	49.9%	32.5%
<i>Constraint</i>	—	—	—	> 199	< 97.2	—	—	—

8.3 Option A — Fuel Economy ($\alpha = 5$, $\pi_c = 40$, $\pi_f = 1.8$)

$$F/\dot{m}_0 = 217 \text{ N/(kg/s)} \quad \text{TSFC} = 14.2 \text{ mg/(N·s)} \quad \eta_T = 69.5\% \quad \eta_P = 59.2\% \\ \eta_O = 41.1\%$$

- Highest η_O and lowest TSFC of all three options.
- $\pi_c = 40$ pushes thermal efficiency to its practical maximum; $\alpha = 5$ is the highest feasible value at this π_c and π_f .
- Specific thrust (217 N/(kg/s)) exceeds $ST_{\min} = 199$ by only 9% — adequate but with limited margin for off-design (hot-and-high takeoff, single-engine climb).
- Best suited to the G650's primary mission: ultra-long-range cruise where fuel burn dominates operating cost.
- Trade-off: highest π_c demands the most advanced compressor aerodynamics and highest material strength. Manufacturing complexity is greatest.

8.4 Option B — Balanced ($\alpha = 4$, $\pi_c = 30$, $\pi_f = 1.7$)

$$F/\dot{m}_0 = 240 \text{ N/(kg/s)} \quad \text{TSFC} = 16.4 \text{ mg/(N}\cdot\text{s)} \quad \eta_T = 66.9\% \quad \eta_P = 53.0\% \\ \eta_O = 35.5\%$$

- 21% thrust margin above $ST_{\min}$ — much better off-design flexibility than Option A.
- $\pi_c = 30$ is a well-established pressure ratio for modern turbofan engines and represents mature, lower-risk technology.
- TSFC is 15% higher than Option A, but still 83% below the mission budget constraint.
- This is the most practical design: it resembles the pressure ratios seen in engines of comparable thrust class and represents a good engineering compromise between performance and complexity.
- **Recommended as the primary design candidate.**

8.5 Option C — Thrust Margin ($\alpha = 3$, $\pi_c = 25$, $\pi_f = 1.8$)

$$F/\dot{m}_0 = 286 \text{ N/(kg/s)} \quad \text{TSFC} = 17.9 \text{ mg/(N}\cdot\text{s)} \quad \eta_T = 65.2\% \quad \eta_P = 49.9\% \\ \eta_O = 32.5\%$$

- Highest specific thrust (286 N/(kg/s)) — 44% above $ST_{\min}$, providing the most robust thrust margin for single-engine operations, obstacle clearance, and ETOPS scenarios.
- Lowest bypass ratio means a physically smaller, lighter fan — advantageous if nacelle diameter is constrained.
- TSFC is the highest of the three options but still only 18.4% of the mission allowance. Long-range cruise performance is not significantly compromised.
- Propulsive efficiency is lowest (49.9%) because lower α means a higher jet velocity relative to flight speed — wasting kinetic energy.
- Suitable if the design requirement is revised to include higher-altitude single-engine segment or stringent climb gradient.

9 Parametric Trends

The following trends are observed from the carpet plot and are consistent with the ideal turbofan theory :

1. **Effect of bypass ratio α :** Increasing α at fixed π_c and π_f *decreases* specific thrust (more total mass flow, lower jet velocity differential) and *decreases* TSFC (more propulsive work from the fan at higher efficiency). The carpet curves shift towards lower TSFC and lower $F/\dot{m}_0$ as α increases.
2. **Effect of compressor pressure ratio π_c :** Increasing π_c along a given α curve moves the operating point towards higher $F/\dot{m}_0$ (up to a plateau) and lower TSFC. The thermal efficiency $\eta_T = 1 - 1/(\tau_r\tau_c)$ increases monotonically with π_c . Cross-marks from $\pi_c = 15$ to 30 show this leftward-downward trend.
3. **Effect of fan pressure ratio π_f :** For fixed α and π_c , increasing π_f raises V_{19} and increases thrust from the bypass stream, but the turbine must extract more power (increasing τ_t work). There is an optimum π_f (Section 5.11 of Mattingly) that maximises $F/\dot{m}_0$ and minimises TSFC simultaneously — corresponding to $V_9 = V_{19}$ and $FR = 1$.
4. **Thermal vs. propulsive efficiency trade-off:** High- α engines sacrifice thermal efficiency slightly (through higher turbine loading) but gain significantly in propulsive efficiency, resulting in better overall efficiency at subsonic cruise — the primary operating condition for the G650.

10 Station Numbering

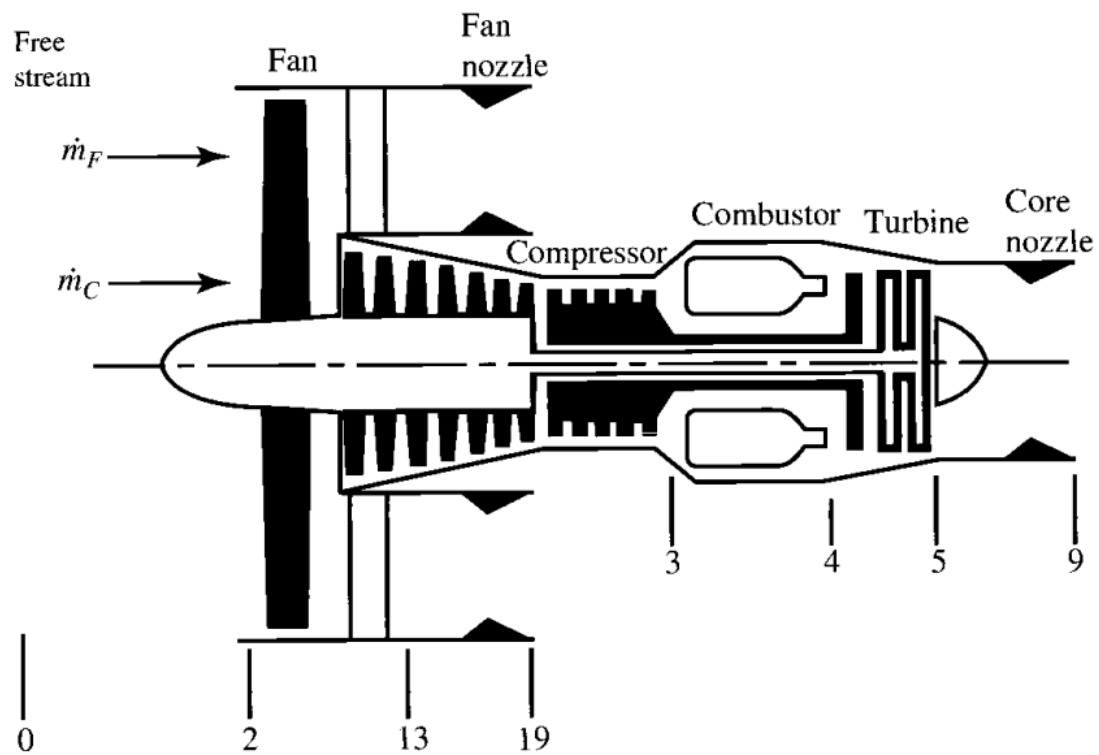


Figure 3: Ideal dual-stream turbofan engine

11 T-s Diagrams

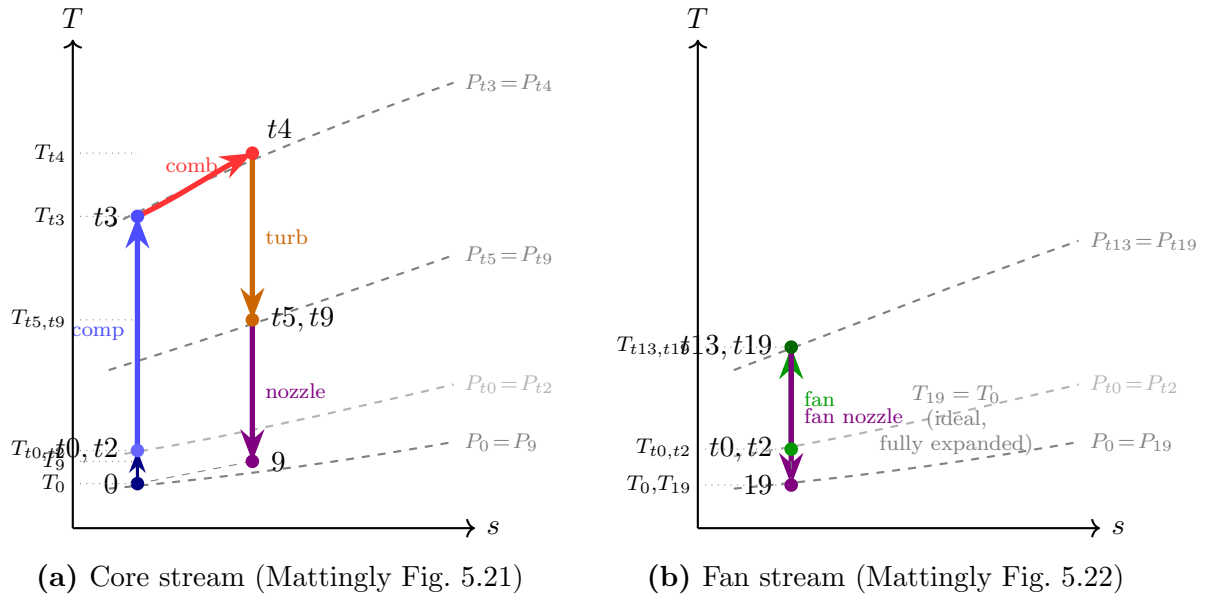


Figure 4: T-s diagrams for the ideal dual-stream turbofan. **Isentropic processes are vertical** (constant entropy s). Combustion ($t3 \rightarrow t4$) is isobaric — moves *rightward* along the $P_{t3} = P_{t4}$ isobar. Both nozzles are fully expanded: $P_0 = P_9 = P_{19}$, so $T_{19} = T_0$ for the fan stream.

12 Summary and Conclusion

Table 3: Summary of key Part 3 results

Item	Value / Finding
Speed of sound a_0	295.0 m/s
Ram ratio τ_r	1.1445
Thermal limit τ_λ	6.924
Max inlet flow (per engine)	59.2 kg/s
$ST_{\min}$ (vertical constraint)	199 N/(kg/s)
$S_{\max}$ (horizontal constraint)	97.2 mg/(N · s)
Feasible α range	1.0 to 7.0 (above 7: $ST < ST_{\min}$ regardless of π_c, π_f)
Feasible π_c range	10 to 40 (upper limit: manufacturing, not thermodynamics)
Feasible π_f range	1.1 to ≈ 2.0 (upper limit: fan tip supersonic speed)
Total feasible designs found	4,751 (from grid scan)
Binding constraint	$ST_{\min}$ (not $S_{\max}$)
Option A ($\alpha = 5, \pi_c = 40, \pi_f = 1.8$)	$ST = 217$, TSFC = 14.2, $\eta_O = 41.1\%$ — <i>recommended for fuel economy</i>
Option B ($\alpha = 4, \pi_c = 30, \pi_f = 1.7$)	$ST = 240$, TSFC = 16.4, $\eta_O = 35.5\%$ — <i>recommended as primary / baseline</i>
Option C ($\alpha = 3, \pi_c = 25, \pi_f = 1.8$)	$ST = 286$, TSFC = 17.9, $\eta_O = 32.5\%$ — <i>recommended for thrust margin</i>

The parametric analysis reveals that the Gulfstream G650 engine design space is fundamentally **thrust-constrained**, not fuel-constrained. All feasible designs are far below the TSFC budget; the critical boundary is the specific thrust floor imposed by the fixed inlet diameter and cruise thrust requirement. This limits the bypass ratio to $\alpha \leq 7$ at the given turbine inlet temperature of 1500 K.

Option B ($\alpha = 4, \pi_c = 30, \pi_f = 1.7$) is recommended as the primary candidate, providing a balanced mix of fuel efficiency, thrust margin, and engineering tractability. It will form the baseline for Part 4 (engine performance analysis) and Part 5 (component design).

12 References

1. Mattingly, J. D., *Elements of Propulsion: Gas Turbines and Rockets*, 2nd ed., AIAA Education Series, 2006. Sections 5.9–5.11, Equations 5.46–5.69.

2. Gulfstream Aerospace, *G650 Aircraft Characteristics*, 2023.
3. International Civil Aviation Organisation (ICAO), *Manual of the ICAO Standard Atmosphere*, 3rd ed., 1993.
4. Group 8, *ME 643 Part 1: Request for Proposal — Gulfstream G650*, IIT Gandhinagar, 2026.
5. Group 8, *ME 643 Part 2: Mission Analysis — Gulfstream G650*, IIT Gandhinagar, 2026.